Computo

Redefining academic publishing through native computational reproducibility

https://computo-journal.org

Julien Chiquet[1] — on behalf of the Computo Editorial Board

[1] Université Paris-Saclay, AgroParisTech, INRAE

September 1, 2026

The problem



Methodological research asserts more than it demonstrates

A methodological paper claims that a procedure works, on the evidence of simulation studies and/or benchmarks — that is, **a computation**.

However, the reader receives a PDF with plots/tables, **not the path that produced them**.

- code and data, when shared at all, are rarely runnable as published
- peer review examines the argument, essentially never the computation
- a year later, the environment has moved and nothing builds

> **!** The premise of Computo
>
> If the computation is the evidence, then it should also be reviewed — and it should still run in ten years.

Computo in one slide

- Created in **2021** under the auspices of the **French Statistical Society** (SFdS)
- **Diamond open access**: free to read, free to publish — ISSN 2824-7795
- **CC-BY 4.0**, authors retain copyright; DOIs registered with DataCite
- Scope: computational and algorithmic contributions in **statistics and machine learning**
- Open peer review on **OpenReview**; partner of **PCI Statistics and ML**
- **23 papers published** since April 2022



https://computo-journal.org
https://github.com/computorg

What we publish

- **Methods papers** — original stats/ML developments, or numerical studies that illustrate theoretical results
- **Case studies and surveys** — including **neutral comparison studies**: when, how and why compared methods perform well or less well
- **Software and tutorial papers** — an implementation *plus* a concrete use case, with benchmarking against related work

> **Primary condition**
>
> Numerical reproducibility is a **necessary condition for publication** (neither a badge nor a supplementary file).

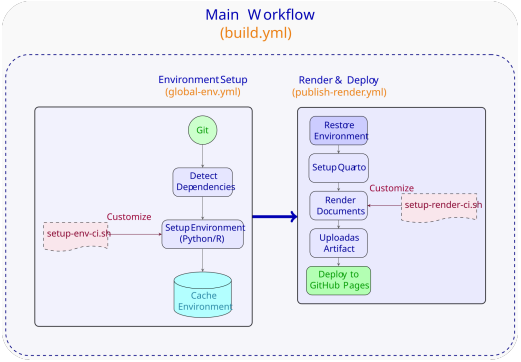
How it works



The stack

Every submission is a **git repository** that builds itself.

1. **Literate programming** — Quarto: text, equations, code and figures in one source
2. **Pinned environment** — `renv.lock`, `requirements.txt`, `Project.toml`
3. **Continuous integration** — rebuilt from scratch on our infrastructure, at every push
4. **Data** — shipped or deposited (e.g. Zenodo)



Templates for **R**, **Python** and **Julia**; the workflow itself is language-agnostic.

Two review tracks, running in parallel

Scientific review

- Reports are public on OpenReview after acceptance
- Reviewers may stay anonymous
- Papers recommended by *PCI Statistics and ML* are accepted without re-review

Reproducibility check

- Automated and continuous, from the first submission
- Run by the editorial/tech team, not the referees
- A paper that does not build is not reviewable

External recognition

A 2026 comparative review of reproducibility policies [1] ranks Computo among the **five journals out of nine** that perform an *independent check* (active, execution-based review) of results, alongside JSS, JASA, Meta-Psychology and the Biometrical Journal.

After acceptance: keep reproducibility

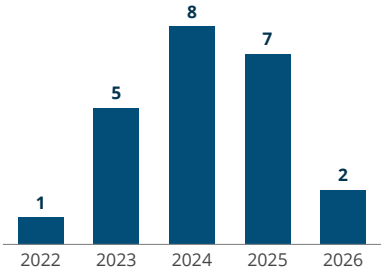
- Article repository transferred to `github.com/computorg/`
- DOI registered on DataCite; metadata should make it indexable
- Review reports published alongside the paper
- Repository archived on **Software Heritage**
- Workflows of published articles are **re-run periodically** to detect decay

> **Long-term reproducibility is a maintenance activity**
>
> Publishing a reproducible artifact is a one-off event. Keeping the article reproducible requires considerably more effort.

Four years of evidence



What the record shows



papers per year *(2026: as of August)*

23 papers, April 2022 to July 2026

Languages actually used:

- **R** only — 14 papers
- **Python** only — 6 papers
- **both** — 3 papers
- **Julia** — template ready, no submission yet

We have shown the feasibility of our policy: submissions do build, and keep building.

What it costs, and what transfers to other fields

The cost falls on the *first* submission: templates absorb most of the technical burden, and editors gain a new form of check.

Still hard

- **Finding authors and reviewers**
- Getting indexed by academic databases
- Heavy computation and large data
- Managing outdated dependencies

What transfers

- Nothing in the stack is specific to statistics: only git, notebook templates, a lockfile and CI
- The editorial move that makes the build a condition is the hard part, but it is “free”
- Separating *scientific* review from the *reproducibility* check keeps referee load unchanged
- Diamond needs light infrastructure, a learned-society host and free software

Come and try it

We are looking for

- **Authors** — methods, benchmarks, software papers
- **Reviewers** — as in any other journal
- **Neutral comparison studies**, which fit the scope exactly

Pre-submission enquiries are welcome: send a title and abstract.

https://computo-journal.org
contact@computo-journal.org
https://github.com/computorg



References

1. **Hornung**, **R**, **Németh**, **L**, **Zadorozhnyi**, **O**, **Ullmann**, **T**, **Kammer**, **M**, **Killick**, **R**, **Paciorek**, **C**, **Chiquet**, **J**, **Herrmann**, **M**, **Batinović**, **L**, **Carlsson**, **R**, **Neuvial**, **P**, **Hejblum**, **B**, **Wrobel**, **J**, **Boulesteix**, **A-L**, and **Tabelow**, **K** 2026 Overcoming barriers to computational reproducibility. *Royal Society Open Science*, 13(6): 252489. DOI: https://doi.org/10.1098/rsos.252489